- Sing-index

$$Y = \theta D + f(X) + \varepsilon$$

(θ: para, D: $\mathbb{D}\theta$, $f(X)$: non)

$$Y = f(\beta^T X) + \varepsilon$$

(f: non, β: para, $\beta^T X$: index)

- Multiple-index

$$Y = f(B^T X) + \varepsilon$$

(B: $p \times d$ matrix, d indices)

$$\mathbb{E}[\varepsilon | X] = 0$$
$$\mathrm{Var}[\varepsilon | X] = g(B^T X)$$

- Sufficient dimension reduction

$$Y | X \sim Y | B^T X$$

$$\mathbb{E}[Y | X] = \mathbb{E}[Y | B^T X] \quad \mathrm{col}(B)$$ central mean space.

$$Y^k = f^k(\beta^{kT} X) + \varepsilon \qquad \beta^k \sim \beta^{k'}$$

$$\mathbb{E}_n[(Y^k - f^k(\beta^{kT} X^k))^2]$$

$$f^k(u) = \frac{\mathbb{E}_n[Y^k K(\beta^k X^k - \beta^k x)]}{\mathbb{E}[K(\beta^k X^k - \beta^k x)]}$$

$$\mathbb{E}_{\mathcal{L}}\left[\left(\hat{\gamma}^{k} - \frac{\mathbb{E}_{\mathcal{L}}\left[\gamma^{k} K\left(\beta^{k} x^{k} - \beta^{k} \tilde{x}^{k}\right)\right]}{\mathbb{E}_{\mathcal{L}}\left[K\left(\beta^{k} x^{k} - \beta^{k} \tilde{x}^{k}\right)\right]}\right)^{2}\right]$$

equations $\gamma = f(\beta x) + \epsilon$

$$\mathbb{E}_{\mathcal{L}_n}\left[\left\{Y - f(\beta x)\right\}\left\{\nabla f(\beta x)\right\}\left\{x - \mathbb{E}[x \mid \beta x]\right\}\right] = 0$$

$$:= G_n(\beta) \qquad \epsilon \qquad H(\beta)$$

$$\min_{\beta^k, \beta^0} \sum_k G_n(\hat{\beta}^k)\left(\beta^k - \hat{\beta}^k\right) + \frac{1}{2}\left(\beta^k - \hat{\beta}^k\right)^T H(\hat{\beta}^k)\left(\beta^k - \hat{\beta}^k\right)$$

$$+ \lambda \sum_k \left|\beta^k - \beta^0\right|$$

sem-parametric efficiency theory.

Causal Inference. Potential Outcome

x	T	Y	Y_0
		$= Y_0(1-T) + Y_1(T)$	Y_1
x_1	$T=1$	Y_1	
x_2	$T=0$	Y_0	

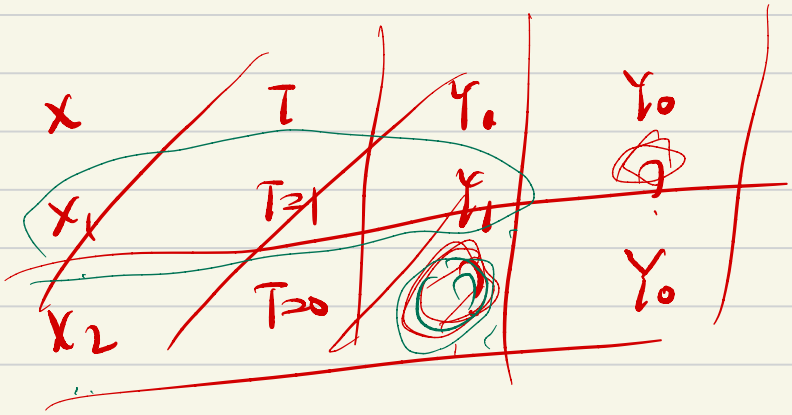
$$ATE = \mathbb{E}[Y_1 - Y_0] \qquad CATE = \mathbb{E}[Y_1 - Y_0 \mid x]$$

$$\mathbb{E}[Y_1 - Y_0 | X] = f(\beta^T X)$$

↑ Federated / Transfer learning.

$$ATE = \mathbb{E}[Y_1 - Y_0]$$

$\mathbb{E}[Y_1]$



Condition?

$T \perp (Y_0, Y_1) | X$

$$\mathbb{E}\left[\frac{Y \mathbb{1}\{T=1\}}{w(x)}\right] \overset{?}{=} \mathbb{E}[Y_1]$$

$w(x) = p(T=1|X)$ ← propensity

IPW ↙

$$= \mathbb{E}\left[\frac{\mathbb{E}[Y \mathbb{1}\{T=1\} | T, X]}{w(x)}\right]$$

$$= \mathbb{E}\left[\frac{\mathbb{E}[T | T=1, X]\, \mathbb{1}\{T=1\}}{w(x)}\right]$$

$$= \mathbb{E}\left[\frac{\mathbb{E}[Y_1 | T=1, X]\, \mathbb{1}\{T=1\}}{w(x)}\right]$$

$\mathbb{E}[Y_1 | T=1, X]$?= $\mathbb{E}[Y_1 | X]$

μ_1

$$= \mathbb{E}[\mathbb{E}[Y | T=1, X]]$$

?

Imputation

DR AIPWE

$$\hat{\theta} = \mathbb{E}_n\left[\frac{\mathbb{1}\{T=1\}}{p(T=1|X)}(Y - \mu_1(X)) + \mu_1(X) - \frac{\mathbb{1}\{T=0\}}{p(T=0|X)}(Y - \mu_0(X)) - \mu_0(X)\right]$$

$\hat{\theta} \sim$?

$$= \mathbb{E}\left[\frac{\mathbb{E}[Y_1|X]\,\mathbb{1}\{T=1\}}{w(x)} \right]$$

$$= \mathbb{E}\left[\mathbb{E}\left[\frac{\mathbb{E}[Y_1|X]\,\mathbb{1}\{T=1\}}{w(x)} \middle| X \right] \right]$$

$$= \mathbb{E}\left[\frac{\mathbb{E}[Y_1|X]}{w(x)} P(T=1|X) \right] \stackrel{?}{=} \mathbb{E}[Y_1]$$

$$w(x) = P(T=1|X) \quad \checkmark$$

$$\frac{T}{\pi_T}\left(Y - \frac{1}{2} T f(\beta x) - m(x) \right) \nabla f(\beta x) \{ x - \mathbb{E}[x|\beta x] \}$$

$$m(x) = (\mu_1 + \mu_0)/2$$

$T = \{1, -1\}$

$$Y = \frac{1}{2} T f(\beta x) + m(x) + \varepsilon \qquad \mathbb{E}[\varepsilon | T, x] = 0$$

$$:= \sum_k G_k(\hat{\beta}^k)(\beta^k - \hat{\beta}^k) + \frac{1}{2}(\beta^k - \hat{\beta}^k)^T H_k(\hat{\beta}^k)(\beta^k - \hat{\beta}^k)$$

$$+ \sum_k \lambda |\beta^k - \beta^0|$$